What the UAE is Doing to Stay Ahead in AI Leadership

The race for artificial intelligence supremacy is reshaping global economic landscapes, and the United Arab Emirates has positioned itself as an unexpected but formidable contender. While traditional technology powerhouses like the United States and China dominate headlines, the UAE has quietly orchestrated a comprehensive, government-backed transformation that offers valuable lessons in strategic AI adoption. This article examines how a nation traditionally known for oil wealth is systematically building an AI-powered future through unprecedented investments, pragmatic partnerships, and a unified national vision.

A Bold National Vision: AI Strategy 2031 and Beyond

The UAE's AI ambitions crystallized in October 2017 when it launched the National AI Strategy 2031, becoming the first nation in the world to appoint a Minister of State for Artificial Intelligence. This wasn't merely symbolic—it signaled a fundamental restructuring of national priorities around artificial intelligence as a cornerstone of economic diversification and future prosperity.

The strategy sets ambitious quantifiable targets that would seem audacious for any nation. By 2031, the UAE aims to generate AED 335 billion (approximately $91 billion) in economic growth from AI, with projections showing AI's contribution to non-oil GDP increasing from approximately 9% to 20%. The government has set a target for AI to represent 13.6% of overall GDP by 2031, creating over 50,000 specialized AI jobs in the process.

This vision rests on eight strategic pillars: building reputation as an AI destination, increasing competitive assets in priority sectors, developing a fertile AI ecosystem, adopting AI across government services, attracting and training talent, bringing world-leading research capabilities, providing essential data infrastructure, and ensuring strong governance frameworks. Each pillar connects specific initiatives to measurable outcomes, creating accountability mechanisms rare in national technology strategies.

Dubai's complementary Dubai 10X initiative exemplifies the strategy's practical implementation. Launched in 2017, this program pushes Dubai's government to operate ten years ahead of other global cities. Results have been tangible—42 innovative projects involving 340 participants and over a million work hours have transformed government operations. The Dubai Electricity and Water Authority (DEWA) alone has implemented AI solutions processing over 2,000 daily screenings, saving customers an estimated AED 17 million annually and conserving 300 million gallons of water through AI-powered leak detection systems.

Unprecedented Financial Commitment: MGX, G42, and Infrastructure Investments

The UAE's AI ambitions are backed by financial commitments that match its rhetoric. In March 2024, the Abu Dhabi government established MGX, a technology investment vehicle designed to manage up to $100 billion in AI-focused assets. Led by Sheikh Tahnoon bin Zayed Al Nahyan, MGX represents a strategic deployment of sovereign wealth that extends far beyond traditional investment approaches.

MGX's partnerships demonstrate the UAE's integration into global AI supply chains. In September 2024, MGX joined forces with Microsoft, BlackRock, and Global Infrastructure Partners to launch the Global AI Infrastructure Investment Partnership (GAIIP), initially targeting $30 billion in private equity with potential to mobilize $100 billion in total investment. These funds are building AI data centers and supporting energy infrastructure primarily in the United States and partner countries, positioning the UAE as a critical facilitator of global AI capacity.

G42, the UAE's sovereign AI champion, stands at the center of this ecosystem. Founded in 2018, G42 has rapidly evolved into a comprehensive AI technology holding group spanning government, healthcare, finance, energy, aviation, and hospitality sectors. Microsoft's $1.5 billion strategic investment in G42 in April 2024 validated its technical capabilities and strategic importance. This partnership has resulted in G42 migrating its data platform to Microsoft Azure, expanding access to advanced AI models, and establishing a $1 billion developer fund to boost regional AI skills.

G42's infrastructure partnerships are equally impressive. Working with Cerebras Systems, G42 built the Condor Galaxy supercomputer cluster in a $100 million deal, deploying systems capable of 4 exaflops of computing power. This infrastructure trained Jais, the world's most advanced Arabic natural language processing model—a 70-billion parameter bilingual system that addresses the critical gap in AI capabilities for Arabic-speaking populations. G42 has deployed 21,500 Nvidia A100 equivalent chips with plans to deliver an additional 60,400 chips, including Nvidia's advanced GB300 series.

The crown jewel of UAE's infrastructure ambitions is Stargate UAE, a collaboration between OpenAI, G42, Oracle, NVIDIA, Cisco, and SoftBank. This next-generation AI infrastructure cluster will run in a newly established 5-gigawatt AI campus in Abu Dhabi—the largest such deployment outside the United States. The first 200-megawatt AI cluster is expected to go live in 2026, with the full 1-gigawatt compute cluster providing capacity equivalent to over 2 million GPUs. Powered by nuclear, solar, and natural gas to minimize carbon emissions, this facility will serve Europe, Africa, and Asia with low-latency connections.

Leading with Implementation: AI Across Key Sectors

Healthcare: Precision Medicine and AI Diagnostics

The UAE's healthcare sector demonstrates how strategic AI deployment creates measurable impact. M42, the nation's tech-enabled healthcare network operating 480 facilities across 27 countries, has pioneered several groundbreaking applications. Med42, M42's open-source Clinical Large Language Model with 70 billion parameters, achieved a 94.5% score on USMLE Sample Exam questions in zero-shot evaluation, outperforming larger closed models like ChatGPT 3.5.

M42's AIRIS-TB system showcases AI's practical impact. This AI-powered imaging tool for tuberculosis screening processes up to 2,000 chest X-rays daily—a tenfold increase over conventional methods. In analyzing over 1 million real-world chest X-rays, AIRIS-TB achieved an AUROC of 98.5% with a 0% false-negative rate, safely automating up to 80% of readings and freeing radiologists for complex cases.

The Emirati Genome Program, with M42 as a key contributor, has collected over 700,000 genetic samples including more than 80,000 nanopore-sequenced samples, creating one of the world's largest long-read cohorts. This initiative has produced approximately 160,000 pharmacogenomics reports, enabling personalized care based on individual genetic profiles and training over 100 Emirati physicians in genomic medicine.

Government Services: Digital-First Transformation

Abu Dhabi's ambition to become the world's first fully AI-native government by 2027 reflects unprecedented digital transformation. The Abu Dhabi Government Digital Strategy 2025-2027 involves AED 13 billion in investment over three years, aiming to automate 100% of government processes and deploy over 200 AI-driven solutions.

Results are already visible. TAMM 3.0, Abu Dhabi's one-stop government services app, has reduced offline customer visits by 90% and made over 73% of transactions instantly available. The strategy has trained over 95% of Abu Dhabi government employees (more than 30,000 individuals) in comprehensive AI programs, establishing Chief Data and AI Officer roles across entities to drive adoption and governance. Economic projections estimate this strategy will contribute over AED 24 billion to Abu Dhabi's GDP by 2027 while creating more than 5,000 new jobs.

Dubai's Universal Blueprint for Artificial Intelligence aims to contribute AED 100 billion annually to Dubai's economy, increasing productivity by 50% through innovative digital solutions. The implementation includes designating a Chief Executive Officer for AI in each government entity, establishing AI and Web3 incubators, and implementing AI week in educational institutions.

Smart Cities: Urban Intelligence at Scale

Dubai and Abu Dhabi exemplify how AI transforms urban living. Dubai's Smart City Strategy has deployed AI-powered infrastructure including intelligent traffic management systems that adapt to real-time conditions and smart energy grids optimizing power distribution. The Roads and Transport Authority (RTA) operates an AI chatbot that has handled over 23 million conversations, while Dubai aims for 25% of all transportation to be autonomous by 2030.

Masdar City in Abu Dhabi serves as a living laboratory for sustainable AI-driven urban development. This carbon-neutral city integrates self-driving electric vehicles, smart grids powered by renewable energy, and AI-optimized building management systems. Dubai Silicon Oasis achieved a 31% reduction in cumulative energy consumption, exceeding its 2030 target years ahead of schedule.

The Sustainable City in DubaiLand demonstrates practical sustainability through solar energy, car-free streets, 10,000 trees, and organic farms—all managed through AI systems that optimize resource consumption. In the Smart Cities Index 2025, Dubai ranked 4th globally and Abu Dhabi 5th, both leading the Middle East and North Africa region.

Finance: Intelligent Banking and Risk Management

UAE banks have embraced AI to enhance customer service, fraud detection, and operational efficiency. By 2027, 85% of banking sector customer interactions in the UAE are expected to be assisted by AI. Emirates NBD's virtual assistant "Eva" provides 24/7 multilingual support with personalized banking recommendations, while AI-powered systems analyze transaction patterns in real-time to identify fraud and money laundering activities.

First Abu Dhabi Bank (FAB) has implemented over 285 automation projects, saving 1.3 million hours of work. Dubai Islamic Bank and Mashreq Bank have invested in AI systems for real-time risk scoring and compliance with anti-money laundering and know-your-customer regulations. The UAE's AI in finance market is projected to grow from $67 million in 2023 to $514 million by 2032, reflecting a compound annual growth rate of 25.3%.

AI is revolutionizing credit scoring by incorporating unconventional data sources like utility payments and social media activity, creating more holistic risk profiles and improving financial inclusion. Banks

are also integrating AI with blockchain technology to enhance transparency and security, particularly for trade finance and identity verification.

Government Support and Progressive Policy Frameworks

The UAE's success stems partly from comprehensive government support structures that balance innovation with responsibility. The UAE Council for Artificial Intelligence, formed by the Cabinet, oversees AI integration across government departments and education, proposes policies for an AI-friendly ecosystem, and promotes public-private sector collaboration.

The Artificial Intelligence and Advanced Technology Council (AIATC) in Abu Dhabi develops financing, investment, and research plans while fostering the emirate as a hub for AI investments and expertise. This council played a pivotal role in establishing MGX and coordinating strategic partnerships with global technology leaders.

Regulatory frameworks prioritize both innovation and ethics. The Federal Personal Data Protection Law No. 45 of 2021 safeguards data privacy while enabling AI development. National Artificial Intelligence Ethics Guidelines provide practical principles for AI developers, emphasizing transparency, accountability, and fairness. The UAE actively participates in international forums to shape global AI governance standards, hosting initiatives like the Global Governance of Artificial Intelligence Roundtable.

The government offers concrete incentives for AI adoption. The AI and coding license allows companies to operate at the DIFC Innovation Hub while offering UAE Golden Visas for employees, attracting international talent. The Dubai AI Campus in the DIFC Innovation Hub represents the largest concentration of AI and advanced technology companies in the MENA region, offering state-of-the-art facilities, incentives, and access to high-performance computing resources including NVIDIA DGX systems.

Strategic International Partnerships: Building Global Bridges

The UAE has masterfully positioned itself as a bridge between Eastern and Western technology ecosystems. Microsoft's total commitment of $15.2 billion to the UAE between 2023 and 2029 includes capital expenses for advanced AI and cloud data centers, local operating expenses, and equity investment in G42. This represents one of Microsoft's largest international commitments, providing the UAE access to cutting-edge AI models from OpenAI, Anthropic, and Microsoft itself.

Microsoft's commitment extends beyond infrastructure to talent development. The company aims to skill one million people in the UAE by the end of 2027, including government employees, students, and teachers. Microsoft established a Global Engineering Development Center in Abu Dhabi and a Microsoft AI for Good Lab, embedding long-term technical capabilities within the nation.

OpenAI's selection of the UAE as the first international location for its Stargate AI infrastructure platform validates the nation's strategic importance. This partnership, involving G42, Oracle, NVIDIA, Cisco, and SoftBank, demonstrates the UAE's ability to convene global technology leaders around shared objectives. The UAE will invest $8 to $10 billion each in Gulf and U.S. Stargate infrastructure, creating reciprocal capacity that benefits both regions.

Academic partnerships complement commercial relationships. The Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), the world's first graduate-level AI research university, collaborates with institutions including École Polytechnique, the University of Chicago, and the Weizmann Institute of

Science. MBZUAI's diverse student body represents over 47 nationalities, and its research focuses on developing AI solutions for underserved regions including weather forecasting for Sub-Saharan Africa and remote healthcare for Pakistan and Indonesia.

The Responsible AI Future Foundation (RAIFF), co-founded by G42, Microsoft, and MBZUAI, exemplifies the UAE's commitment to ethical AI development. The Intergovernmental Assurance Agreement (IGAA) between UAE and U.S. governments ensures compliance with international standards for cybersecurity, export controls, data protection, and responsible AI—a framework that could serve as a model for future international AI cooperation.

The UAE's Unique Formula for AI Success

Several factors distinguish the UAE's approach from other national AI strategies, creating a replicable yet distinctive model for rapid AI adoption.

Unified "Whole-of-State" Vision: Unlike fragmented approaches in many Western democracies or opaque state-controlled models elsewhere, the UAE maintains clear, cohesive direction from the highest levels of government. The seamless alignment between national strategy, ministerial mandates, sovereign investment, academic focus, and infrastructure development minimizes bureaucratic friction and accelerates execution of complex, multi-domain initiatives.

Capital as Strategic Tool: The UAE deploys sovereign wealth not merely for financial returns but as a proactive tool for strategic positioning. By securing stakes in OpenAI, Anthropic, xAI, and other frontier AI companies, the UAE gains influence over technology development direction while embedding itself within critical supply chains. This "capital-as-a-service" approach creates dependencies that benefit long-term strategic objectives.

Rapid Execution Culture: The UAE's governance structure enables decision-making speed impossible in many democratic systems. From appointing the world's first AI Minister in 2017 to launching MGX with $100 billion ambitions in 2024, the nation compresses timelines that would span decades elsewhere. Dubai 10X's mandate to operate ten years ahead of other cities isn't rhetoric—it drives concrete project deliverables with measurable outcomes.

Balanced Global Positioning: The UAE navigates the U.S.-China technology competition by positioning itself as an "indispensable third pole." While deepening partnerships with American technology leaders and complying with U.S. export controls, the nation maintains economic relationships globally. G42's reported divestment from Chinese technology firms to secure Nvidia chip exports demonstrates pragmatic alignment without ideological rigidity.

Energy Advantage: The Middle East's abundant energy resources provide unique advantages for energy-intensive AI infrastructure. The UAE integrates renewable energy (solar and wind) alongside natural gas and nuclear power to create sustainable, cost-effective power for AI campuses. This energy security eliminates constraints facing data center development in energy-constrained regions.

Dual Human Capital Strategy: The UAE simultaneously develops local talent through specialized institutions like MBZUAI while attracting global expertise via Golden Visas and competitive compensation. With 59.4% of the UAE population using generative AI—the highest per capita globally—the nation is building both supply and demand sides of the AI ecosystem.

Lessons for Other Regions and Nations

The UAE's approach offers actionable insights for governments worldwide seeking to accelerate AI adoption while maintaining sovereignty and ethical standards.

Make AI a National Priority with Structural Commitment: Appointing dedicated ministers, creating specialized councils, and establishing clear governance frameworks signals commitment beyond rhetoric. The UAE's institutional architecture ensures AI considerations influence every government decision, from infrastructure investments to education policy.

Deploy Capital Strategically, Not Just Abundantly: While the UAE benefits from sovereign wealth, the strategic deployment matters more than absolute amounts. Smaller nations can create focused investment vehicles targeting specific AI capabilities, securing positions in global value chains without matching UAE's total spending.

Build Enabling Infrastructure Before Demanding Applications: The UAE invested in data centers, supercomputers, cloud platforms, and energy infrastructure before expecting broad AI adoption. This "build it and they will come" approach reduces barriers for entrepreneurs and enterprises, accelerating ecosystem development.

Balance Innovation with Responsible Governance: The UAE's ethical guidelines, data protection laws, and international agreements demonstrate that rapid innovation need not sacrifice responsibility. Early establishment of governance frameworks prevents the regulatory whiplash that constrains innovation in jurisdictions that move slowly then overreact to problems.

Cultivate Global Partnerships Rather Than Autarky: The UAE's success stems partly from recognizing it cannot develop all AI capabilities indigenously. Strategic partnerships with Microsoft, OpenAI, NVIDIA, and others provide access to frontier technologies while UAE's infrastructure, capital, and market access create reciprocal value. This interdependence builds resilient relationships.

Focus on Practical Implementation, Not Just Research: While the UAE invests in fundamental research through MBZUAI, the emphasis on deploying AI in government services, healthcare, and finance creates immediate value. This pragmatic focus attracts talent and investment by demonstrating real-world impact rather than theoretical potential.

Create Sandboxes for Experimentation: The DIFC Innovation Hub, Dubai 10X projects, and Masdar City provide controlled environments for testing AI applications before broader deployment. These sandboxes generate lessons that inform policy while limiting downside risks from failures.

Conclusion: A Blueprint for the AI Era

The UAE's transformation from oil-dependent economy to AI powerhouse offers a masterclass in strategic national development. Within less than a decade, the nation has built comprehensive AI infrastructure, secured partnerships with every major global technology company, trained tens of thousands in AI skills, and deployed AI across government services affecting millions of residents.

The results speak to the strategy's effectiveness. Dubai and Abu Dhabi rank among the world's top five smart cities. The UAE leads globally in per capita generative AI usage. Investments like MGX's $100 billion AI fund and Stargate UAE's 5-gigawatt data center position the nation as a critical node in global AI infrastructure. Economic projections show AI contributing $91 billion annually by 2031, fundamentally reshaping the national economy.

What makes the UAE's approach particularly valuable for other nations is its replicability despite unique circumstances. While sovereign wealth provides advantages, the core principles—unified national vision, rapid execution, strategic global partnerships, practical implementation focus, and balanced governance—are accessible to nations regardless of size or resource endowment.

As artificial intelligence reshapes global economic and geopolitical landscapes, the UAE's experience demonstrates that newcomers can achieve leadership positions through strategic planning, decisive action, and pragmatic partnerships. The nation's journey from AI strategy announcement in 2017 to global AI infrastructure leader by 2024 suggests that in the AI era, strategic agility and execution speed matter more than historical advantages. For governments worldwide seeking to position their nations for an AI-powered future, the UAE provides not just inspiration but a detailed roadmap for transformative change.

References

UAE National Strategy for Artificial Intelligence 2031 - Digital Watch (https://dig.watch/resource/the-uae-national-strategy-for-artificial-intelligence-2031)

UAE National AI Strategy 2031 and Program - UNESCO (https://www.unesco.org/creativity/en/policy-monitoring-platform/national-ai-strategy-2031-and-national-program-artificial-intelligence)

UAE Strategy for Artificial Intelligence - U.AE (https://u.ae/en/about-the-uae/strategies-initiatives-and-awards/strategies-plans-and-visions/government-services-and-digital-transformation/uae-strategy-for-artificial-intelligence)

UAE AI Strategy 2031: Global Leader - The SOO Group (https://thesoogroup.com/blog/uae-ai-strategy-2031-global-leader)

Inside UAE and Dubai's Bold AI Strategy - Digital Bricks (https://www.digitalbricks.ai/blog-posts/inside-uae-and-dubais-bold-ai-strategy)

DEWA's Adoption of Generative AI - Dubai Electricity and Water Authority (https://www.dewa.gov.ae/en/about-us/media-publications/latest-news/2024/02/dewas-adoption-of-generative-ai)

Dubai 10X Initiative - Dubai Future Foundation (https://www.dubaifuture.ae/initiatives/future-design-and-acceleration/dubai-10x)

AI Enables DEWA to Innovate - Dubai Electricity and Water Authority (https://www.dewa.gov.ae/en/about-us/media-publications/latest-news/2022/08/ai-enables-dewa-to-innovate)

The United Arab Emirates' AI Ambitions - CSIS (https://www.csis.org/analysis/united-arab-emirates-ai-ambitions)

UAE AI Strategy Analysis for Global AI Dominance - Klover.ai

MGX Joins GAIIP Consortium Data Centres - Middle East AI News (https://www.middleeastainews.com/p/mgx-joins-gaiip-consortium-data-centres)

BlackRock, Global Infrastructure Partners, Microsoft, and MGX Launch AI Partnership - Microsoft News (https://news.microsoft.com/source/2024/09/17/blackrock-global-infrastructure-partners-microsoft-and-mgx-launch-new-ai-partnership-to-invest-in-data-centers-and-supporting-power-infrastructure/)

MGX Fund Management Limited - Wikipedia (https://en.wikipedia.org/wiki/MGX_Fund_Management_Limited)

G42 - Official Website (https://www.g42.ai/)

G42 and Cisco Extend Strategic Partnership - Cisco Newsroom (https://newsroom.cisco.com/c/r/newsroom/en/us/a/y2025/m05/g42-and-cisco-extend-strategic-partnership-to-drive-ai-innovation-and-infrastructure-development.html)

Microsoft and G42 Partner to Accelerate AI Innovation - Microsoft Blog (https://blogs.microsoft.com/blog/2024/04/15/microsoft-and-g42-partner-to-accelerate-ai-innovation-in-uae-and-beyond/)

Global Tech Alliance Launches Stargate UAE - G42 (https://www.g42.ai/resources/news/global-tech-alliance-launches-stargate-uae)

M42 Announces New Clinical LLM - G42 (https://www.g42.ai/resources/news/m42-announces-new-clinical-llm-transform-future-ai-healthcare)

UAE's AI Healthcare: The Future of Medicine - Gulf Business (https://gulfbusiness.com/uaes-ai-healthcare-the-future-of-medicine/)

How M42 is Redefining Healthcare Through AI - Gulf Business (https://gulfbusiness.com/how-m42-is-redefining-healthcare-through-ai/)

M42 - Official Website (https://m42.ae/)

AI-Driven Smart Cities: Dubai and Abu Dhabi - Go-Globe (https://www.go-globe.com/ai-driven-smart-cities-dubai-and-abu-dhabi/)

Smart Sustainable Cities - U.AE (https://u.ae/en/about-the-uae/digital-uae/digital-cities/smart-sustainable-cities)

How the UAE is Using AI and Smart Tech - UAE Stories (https://uaestories.com/how-the-uae-is-using-ai-and-smart-tech-to-build-the/)

AI-Powered Smart Cities: How the UAE is Leading - UAE Stories (https://uaestories.com/ai-powered-smart-cities-how-the-uae-is/)

UAE Smart Cities: Leading Digital Transformation - Interdev Technology (https://www.interdevtechnology.com/post/uae-smart-cities-leading-the-future-of-digital-transformation)

AI-Enabled Digital Transformation via PPPP - U.AE (https://u.ae/en/information-and-services/business/public-private-people-partnership/pppp/ai-enabled-digital-transformation-via-pppp)

Dubai Universal Blueprint for AI - U.AE (https://u.ae/en/about-the-uae/strategies-initiatives-and-awards/strategies-plans-and-visions/government-services-and-digital-transformation/dubai-universal-blueprint-for-ai)

Abu Dhabi Set to Become First Fully AI-Powered Government - CIO (https://www.cio.com/article/3812023/abu-dhabi-set-to-become-the-worlds-first-fully-ai-powered-government-by-2027.html)

Artificial Intelligence in Government Policies - U.AE (https://u.ae/en/about-the-uae/digital-uae/digital-technology/artificial-intelligence/artificial-intelligence-in-government-policies)

Banking and AI Developments in the UAE - Hadef & Partners (https://hadefpartners.com/news-insights/insights/banking-and-ai-developments-in-the-uae-paving-the-way-for-innovation/)

The Role of AI in Dubai’s Financial Sector - Alpha Partners (https://www.alphapartners.co/blog/the-role-of-ai-in-dubais-financial-sector-trends-and-future-outlook)

UAE Artificial Intelligence in Finance Market - Credence Research (https://www.credenceresearch.com/report/uae-artificial-intelligence-in-finance-market)

Embracing AI to Unlock the Future of Banking - Gulf Business (https://gulfbusiness.com/embracing-ai-to-unlock-the-future-of-banking/)

Microsoft’s $15.2 Billion Investment in the UAE - Microsoft Blog (https://blogs.microsoft.com/on-the-issues/2025/11/03/microsofts-15-2-billion-usd-investment-in-the-uae/)

Introducing Stargate UAE - OpenAI

UAE-OpenAI Stargate Deal - Axios

In-Depth Analysis of UAE’s AI Strategy - Marketing Shots (https://marketing-shots.com/digital-transformation/uae-artificial-intelligence-strategy/)

AI UAE: Paving the Way to Become a Global AI Leader - Virtuzone (https://virtuzone.com/blog/ai-uae/)

MBZUAI Building Humanity’s Symbiotic Relationship with AI - Forbes

Mohamed bin Zayed University of Artificial Intelligence - Wikipedia (https://en.wikipedia.org/wiki/Mohamed_bin_Zayed_University_of_Artificial_Intelligence)

L’X and MBZUAI Sign MoU Focusing AI - École Polytechnique (https://www.polytechnique.edu/en/news/lx-and-mbzuai-sign-mou-focusing-ia)

MBZUAI Highlights AI Innovation in Energy Sector - EdTech Innovation Hub (https://www.edtechinnovationhub.com/news/mbzuai-highlights-ai-innovation-in-energy-sector-at-adipec-2025)